

LINGYU(Alvin) XU

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Education

Worcester Polytechnic Institute, Worcester, MA

Jan 2024 - Dec 2025

Master of Science in Information Technology in System Design & Artificial Intelligence

- GPA: 3.75
- **Coursework:** Application Development, Machine Learning, Data Mining, Data Visualization, SQL

Pennsylvania State University, State College, PA

Aug 2015 - May 2020

Bachelor of Science in Computer Science

- **Coursework:** System Programming, Network Programming, Algorithm, Data Structure, Statistics

Skills

Languages: Python, Golang, Java, Angular, JavaScript, TypeScript, HTML5, CSS, C, C++

Databases & Data Analysis: MySQL, PostgreSQL, Tableau, Excel

Data Science & Machine Learning: TensorFlow, PyTorch, Scikit-Learn, Pandas, Spark, Scala, PIG

DevOps & CI/CD: Docker, Kubernetes, Jenkins, PACT, Git

Tools & Project Management: REST APIs, Agile, JIRA, Confluence, Mendix Development

Experience

Research Assistant

Worcester, MA

Worcester Polytechnic Institute

May 2024 - July 2024

- Reviewed and discussed blockchain research papers to design innovative financial tracking ideas.
- Containerized 2 local services to enhance scalability, stability, and deployment efficiency with 20%.

Full-Stack Software Development Engineer

Shanghai, China

Dell Technologies

Mar 2021 - Dec 2023

- **Front-End Development:** Enhanced user experience by designing intuitive UI components (filters, forms, dropdowns) for initial setup, improving interaction and data presentation. Achieved over 93% test coverage through diligent contract and unit testing of modules.
- **Back-End Development:** Improved server deployment reliability by 10% through the introduction of pre-validation steps and authored APIs to support server clustering. Established a robust network and storage environment for both Windows and Linux servers. Developed a SQL database for managing server configurations and user credentials, ensuring thorough testing with over 80% coverage.
- **Automation & Testing:** Spearheaded the development of automated testing pipelines for cluster deployments, accommodating 6 configurations, which enhanced adaptability and efficiency. Proactively maintained CI/CD pipelines, addressing bugs and technical issues to streamline processes.

Software Development Engineer Intern(Front-end)

Shanghai, China

ZF Group

May 2018 - Aug 2018

- Engineered a document management system with SharePoint, customizing interfaces for 12 different departments to facilitate a paperless office environment and an access-based file sharing platform.
- Modified CSS, HTML, and JavaScript to enhance visual appeal and usability; optimized layout with business brand identity, improving user experience and reducing process time by 15%.

Projects

Android App Development, Worcester, MA

- **Application Development:** Built an Android app with 6 pages in Java & JavaScript, simulating banking operations like balance checks, transaction filtering, and sorting with a local SQLite database.
- **Database Design:** Designed and implemented database schema with 6 tables, data generation with Python, and backend processing logic with Java for efficient data storage and retrieval.
- Tools Used: Java, Python, JavaScript, SQL

Intelligent Private Banking Portal, Capital One, State college, PA

- **Machine Learning:** Developed predictive models with TensorFlow using linear regression and decision tree methodologies to forecast monthly net and category-specific spending with 83% accuracy.
- **Integration:** Integrated predictive analytics with a user-friendly visual interface on personal online banking website.
- **Reward:** Prize at the 2019 Capstone Design Showcase
- Tools Used: TensorFlow, Python